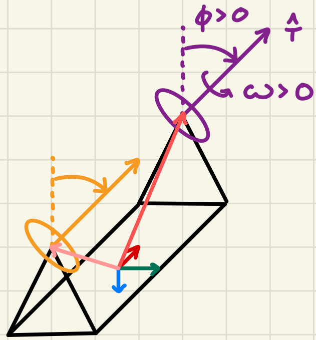
Rotor As wheel Derivation



$H_{rotor} = \hat{T}\omega j$

$\dot{H}_{rotor} = \frac{d}{dt}_B(\hat{T}\omega j) + \hat{T}\omega j \times {}^I\omega^B$

$\dot{\hat{T}}\omega j + \hat{T}\dot{\omega} j + \hat{T}\omega j \times {}^I\omega^B$

$M_{\text{on rotor}} = j(\dot{\hat{T}}\omega + \hat{T}\dot{\omega} + \hat{T}\omega \times {}^I\omega^B)$

$\therefore M_{\text{on body}} = -j(\dot{\hat{T}}\omega + \hat{T}\dot{\omega} + \hat{T}\omega \times {}^I\omega^B)$

$\hat{T} = \begin{bmatrix} 0 \\ S\phi \\ -C\phi \end{bmatrix} \quad \therefore \dot{\hat{T}} = \begin{bmatrix} 0 \\ S\dot{\phi} \\ -C\dot{\phi} \end{bmatrix}$

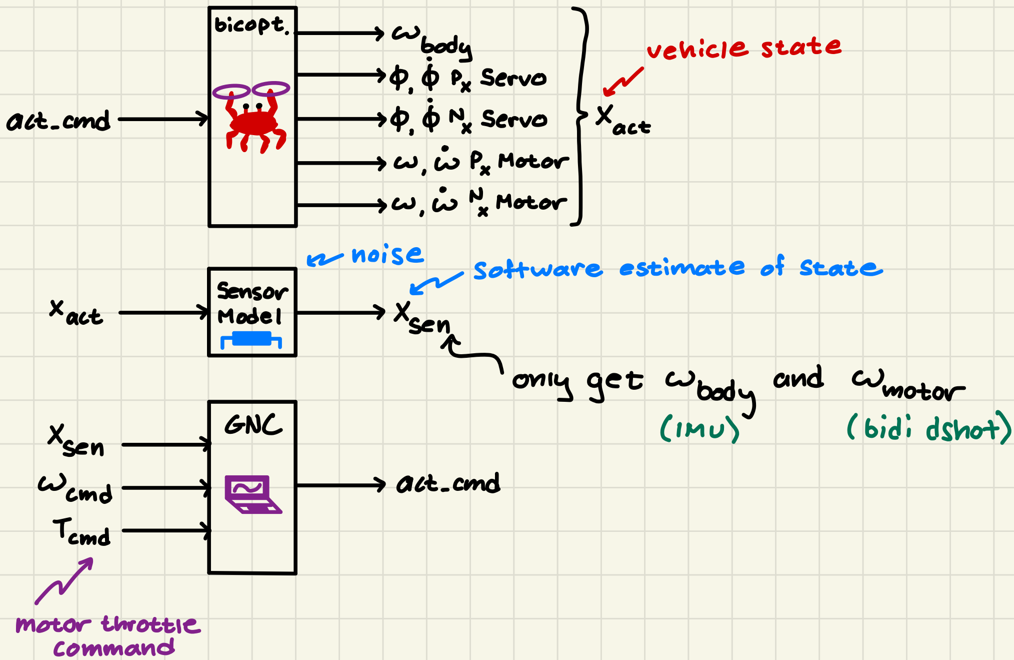
Assumptions

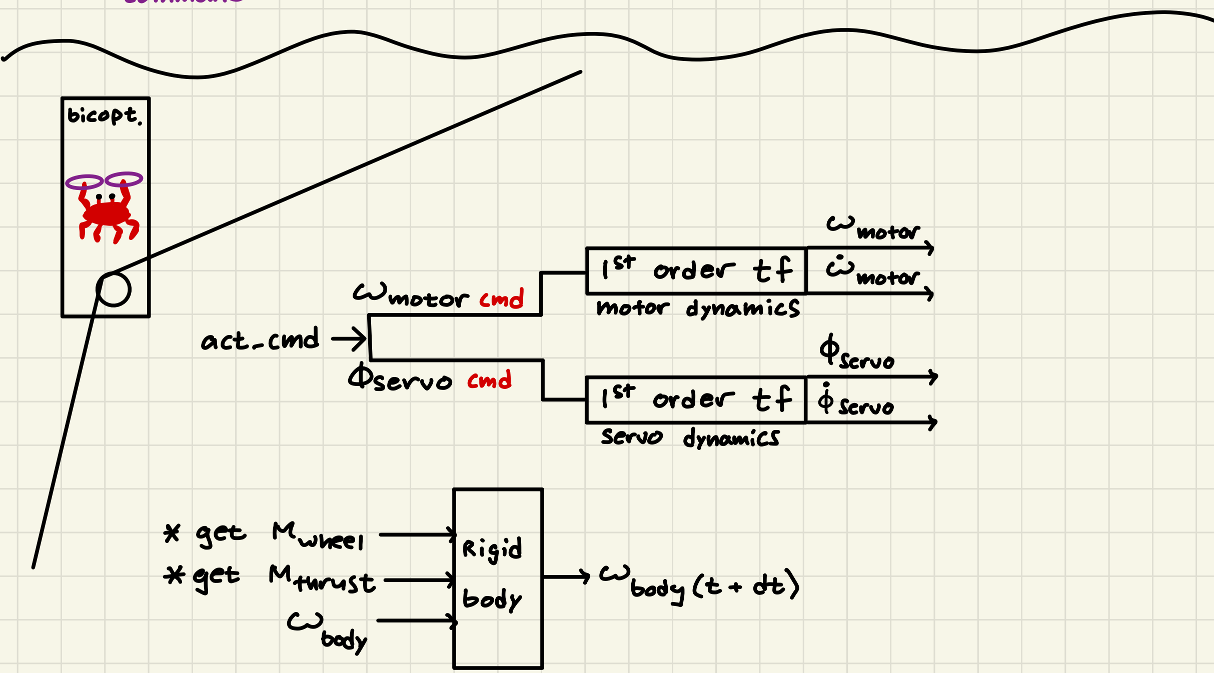
ϕ is only affected by the servos

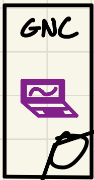
ω is only affected by the motors

$H_{tilt} << H_{rotor}$

SIM Architecture planning







nonlinear

$\omega_{b,sen} \rightarrow$ low-pass filter $\rightarrow \omega_{b,est}$

$\xrightarrow{\omega_{M,sen}} = \xrightarrow{\omega_{M,est}}$

we want to estimate this

$$\boxed{M_{on\ body} = -j\left(\dot{\hat{T}}\omega + \hat{T}\dot{\omega} + \hat{T}\omega \times {}^{I}\omega^{B}\right)}$$

but,

$M_{on\ body}\left(\phi, \dot{\phi}, \omega_M, \dot{\omega}_M, {}^{I}\omega^{B}\right)$

and

$M_{thrust}\left(\phi, \omega, {}^{I}\omega^{B}\right)$

because the command is a function of $E = \omega_{B,des} - \omega_{B,est}$

$\therefore$ coupling & instability

Idea: If we can predict what M_{wheel} will be and pre-compensate, the loop is broken. (feed fwd)

Need:

$$\left.\begin{array}{l} \omega_b(t+dt) \\ \omega_m, \dot{\omega}_m(t+dt) \\ \phi, \dot{\phi}(t+dt) \end{array}\right\} \text{as a function of } u \text{ and current state}$$

actuator command

1) generate u as a function of $\omega_{b,est}$ and $\omega_{b,des}$

estimate this

$$\dot{\omega} = \omega\,[A] + bu$$

Question: given what we know about the dynamics, can we help the estimator out by giving it A (unknowns) and only estimate unknowns rather than a whole 3x3 matrix?

Problem:

in reality...

$${}^{I}\dot{\omega}^{B} = A\,{}^{I}\omega^{B} + Bu + u_{dist}$$

$B = \begin{bmatrix} \frac{1}{I_1} \\ \frac{1}{I_2} \end{bmatrix}$

A: fcn of $I_1, I_2 I_3$ = fcn of $({}^{I}\omega^{B}, I_1$

$$I_1\,\dot{\omega}_1 + (I_3 - I_2)\,\omega_2\,\omega_3 = M_1$$
$$I_2\,\dot{\omega}_2 + (I_1 - I_3)\,\omega_3\,\omega_1 = M_2$$
$$I_3\,\dot{\omega}_3 + (I_2 - I_1)\,\omega_1\,\omega_2 = M_3$$

$\rightarrow \dfrac{\cancel{M} - (I_3 - I_2)\omega_2\omega_3}{I_1}$ ignoring

But maybe we can use this!

$$u_{dist} = -J\left(\hat{T}\omega_M + \hat{T}\dot{\omega}_M + \hat{T}\omega_M \times {}^{I}\omega^{B}\right)$$

have a direct estimate

function of direct estimate

guess based on inputs

TLDR:

* inertias (constant)
* vehicle state (changing)

but...

$f = -(A\,{}^{I}\omega^{B} + u_{dist})$ but we don't care

$${}^{I}\dot{\omega}^{B} = Bu - f \quad \text{(Adrc)}$$

$$f = Bu - {}^{I}\dot{\omega}^{B} \rightarrow f + {}^{I}\dot{\omega}^{B} = Bu \qquad B^T(f + {}^{I}\dot{\omega}^{B})$$

$$u = ({}^{I}\dot{\omega}^{B} + f)B^T$$

basically just your control gains

1) estimate f

$${}^{I}\dot{\omega}^{B}_{prev} = \big(\underbrace{{}^{I}\omega^{B}_{EST,t-1} - {}^{I}\omega^{B}_{EST,t-2}}_{\text{yucky numerical differentiation}}\big)/dt\_s$$

$$f = B\,u_{prev} - {}^{I}\dot{\omega}^{B}_{prev}$$

2) compute u

$${}^{I}\dot{\omega}^{B}_{curr} = \left({}^{I}\omega^{B}_{EST,t-0} - {}^{I}\omega^{B}_{EST,t-1}\right)/dt\_s$$

$$u = B^T(f + {}^{I}\dot{\omega}^{B}_{curr})$$

new idea: integrate everything

Question: how does the system evolve?
Answer:

$$ {}^{I}\dot{\omega}^{B} = \underset{\text{input}}{U} + \underset{\text{disturbance}}{\dot{f}} $$

what is f?

$$ {}^{I}\omega^{B} = \int U + f $$
$$ \therefore f = {}^{I}\omega^{B} - \int U $$

ok; what if I want ${}^{I}\omega^{B}$ to be ${}^{I}\omega^{B}$

$$ {}^{I}\omega^{B} = \int_{0}^{t-1} U + \Delta \int_{t-1}^{t} U + f(t) $$
$$ {}^{I}\omega^{B} - \dot{\int} u - \dot{f}_{est}(t) = \dot{\int} u $$

$$ \therefore U = \underset{\text{numerical derivative of control input}}{{}^{I}\dot{\omega}^{B}} - \dot{f}_{est} $$ ← numerical derivative of generalized disturbance estimate

okay lets uhhhh do it who needs gains

wrong! f is a function of ${}^{I}\omega^{B}$ actual

but that is ~~open loop~~. let's have one more go

I think using ${}^{I}\dot{\omega}^{B}$ (numerical derivative) is ok but it should not be used to get f without some form of inherent numerical stability

1) say ${}^{I}\dot{\omega}^{B} = U + \dot{f}$

let's estimate f the same way we did before

$$ \therefore {}^{I}\omega^{B} - \int U = f $$

2) let ${}^{I}\omega^{B}$ be our desired state

if ${}^{I}\omega^{B}_{actual} = \int U + f$, then

f can be estimated as ${}^{I}\omega^{B}_{actual} - \int U = f$

← note that f is not itself being integrated; but is instead re-computed every time.

so, ${}^{I}\dot{\omega}^{B} = u + \dot{f}$

and

$${}^{I}\omega^{B} = \int u + f$$

we want ${}^{I}\omega^{B} - {}^{I}\omega^{B} = 0$

so,

$$\int u + f - {}^{I}\omega^{B} = 0$$

$$\int u = \int_{0}^{t-1} u + \int_{t-1}^{t} u$$

so, $\int_{0}^{t-1} u + \int_{t-1}^{t} u + f - {}^{I}\omega^{B} = 0$

so, ${}^{I}\omega^{B} =$

wait! haha